

Computer Vision

Milestone 1

Literature Review

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1 Problem Context

The automated analysis of football match video represents a critical frontier in sports analytics and computer vision. While professional teams and broadcasters employ expensive hardware-based tracking systems (such as Prozone), these remain inaccessible to smaller clubs, youth academies, and researchers. The core challenge lies in extracting actionable tactical and performance insights from raw broadcast footage with minimal infrastructure investment.

Football video analysis inherently involves multiple interdependent computer vision problems: detecting and tracking players across frames despite occlusion and fast motion, assigning team membership based on visual appearance, registering camera perspectives to real-world pitch coordinates, and synthesizing tracking data into granular performance metrics and formation patterns. Most existing research addresses these problems in isolation: tracking pipelines often ignore tactical synthesis; formation analysis typically assumes pre-processed tracking data; and calibration frameworks operate independently from downstream analytics tasks.

This project aims to unify these components into an integrated end-to-end pipeline that accepts raw video and produces both individual player performance profiling and real-time team formation mapping, delivered through an intuitive dual-view dashboard. The system is designed for accessibility (using publicly available models and datasets), accuracy (leveraging recent advances in deep learning for detection and tracking), and interpretability (grounding all measurements in real-world pitch coordinates rather than pixel space).

2 Paper Summaries

2.1 Voronoi Analysis of a Soccer Game (Kim)

Objective: This foundational study applies Voronoi diagrams to quantify player and team contributions during a match by partitioning pitch space into control zones and examining spatial dominance patterns.

Methodology: The author utilizes Voronoi partitioning to assign each point on the pitch to the closest player. The concept of “excess Voronoi area” is introduced, calculated by subtracting a player’s nominal formation area from their observed in-match area, thereby measuring relative engagement intensity.

Dataset: Simulated computer soccer game with precise coordinate tracking for each player, enabling controlled analysis of spatial behavior without video processing challenges.

Key Results: Dominant teams sustained area ratios up to 4.3 during goal-scoring moments. Mean polygon vertices per player ranged from 5.2 to 6.1, marginally lower than expected under random placement.

Limitations: Dataset is simulated rather than video-derived, circumventing real-world occlusion and motion blur. Publication predates the five-year recency cutoff, reflecting methodologies less aligned with modern computer vision practice.

2.2 TVCalib: Camera Calibration for Sports Field Registration (Theiner & Ewerth)

Objective: Propose a differentiable framework that recovers 3D camera intrinsics and extrinsics from uncalibrated sports broadcast frames, treating field registration as a camera calibration optimization problem.

Methodology: DeepLabV3 performs semantic segmentation of pitch lines and circles. A differentiable reprojection loss function is defined over 3D primitives (lines, circles). Iterative optimization minimizes this loss, with a self-verification module rejecting frames exceeding a confidence threshold.

Dataset: SoccerNet-Calibration and World Cup 2014 datasets were used for evaluation.

Key Results: Achieved a Compound Score of 76.9 on SN-Calib using ground-truth segmentations, outperforming 2D registration baselines. Multi-view initialization (center, left, right) substantially improved convergence.

Limitations: Iterative optimization is prone to local minima, especially if initial camera distribution is sparse. Performance degrades when segmentation relies on neural network predictions rather than ground truth.

2.3 SportsMOT: A Large Multi-Object Tracking Dataset (Cui et al.)

Objective: Address the scarcity of sports-specific multi-object tracking benchmarks by providing a large-scale dataset with diverse sports and challenging MOT scenarios (fast motion, occlusion, similar appearance).

Methodology: Introduces MixSort, which integrates appearance-based association (MixFormer-like modules) into tracking-by-detection frameworks such as ByteTrack and OC-SORT. Appearance embeddings are target-specific, improving robustness in crowded scenes.

Dataset: 240 sequences spanning basketball, volleyball, and football (150K+ frames, 1.6M+ bounding boxes). Football sequences include multiple camera angles and lighting conditions.

Key Results: MixSort-OC achieved state-of-the-art HOTA of 74.1 and IDF1 of 74.4. Standard Kalman filtering showed poor performance on variable-speed sports motion; simple IoU-based matching often outperformed classic filtering in this domain.

Limitations: Requires specialized post-processing to filter non-player detections (audience, referees). Contact-heavy sports (basketball) remain challenging for appearance-based association despite improvements.

2.4 Enhancing Soccer Camera Calibration Through Keypoint Exploitation (Falaleev & Chen)

Objective: Overcome the scarcity of easy-to-detect calibration markers in soccer broadcasts by systematically exploiting a dense grid of structural keypoints (line intersections, conic tangents, conic intersections).

Methodology: HRNetV2 detects 57 semantic keypoint classes. An “Iterative Voter” algorithm selects the most reliable point subsets based on confidence thresholds optimized through hyperparameter search (Optuna framework).

Dataset: SoccerNet Camera Calibration Challenge 2023, consisting of 25,000+ annotated images spanning diverse broadcast contexts.

Key Results: Won the SoccerNet 2023 Challenge with a final score of 0.5535. Intersection-based keypoints (with clear geometric markings) achieved the highest accuracy; auxiliary tangent points improved coverage but at lower precision.

Limitations: Accuracy drops significantly when forced to rely on tangent points or homography-projected points lacking visible markings. Radial lens distortion correction remains a challenging optimization problem.

2.5 FootyVision: Multi-Object Tracking and Localisation (Andrews et al.)

Objective: Deliver a complete, single-stage system for real-time player and ball tracking, alongside tactical field localization, directly from raw football video.

Methodology: YOLOv7 backbone for object detection. Hungarian algorithm assigns track identities based on appearance embeddings, Euclidean distance, and velocity. Perspective transformation is computed via line and ellipse sampling from activation maps.

Dataset: Evaluated on ISSIA, SoccerNet, and NRK (Norwegian broadcast) datasets with diverse pitch geometries and lighting.

Key Results: Achieved 95.7% mAP for object detection and 94.04% MOTA on SoccerNet test set. Robust to partial occlusions; demonstrated stable tracking through brief player overlap.

Limitations: Real-time performance remains unreliable on standard hardware. Identity switches (IDSW) occur during extended full occlusions. Wide-angle lens distortion causes perspective inaccuracies.

2.6 Team Style Analysis using Spatiotemporal Tracking Data (Bialkowski et al.)

Objective: Characterize unique team behavioral “styles” over an entire season by leveraging multi-agent spatiotemporal tracking patterns and formation descriptors.

Methodology: Linear Discriminant Analysis processes match-level descriptors including ball possession statistics and a novel “formation descriptor” derived from role-occupancy maps. Team identity is then predicted using k -NN classification in the learned feature space.

Dataset: One full professional season of Prozone tracking data spanning 20 teams and 375 matches.

Key Results: Formation descriptor alone achieved 67.32% team identification accuracy. Fusion of all descriptors increased the rate to 70.38%. Formation patterns emerged as the strongest discriminator of team identity.

Limitations: Published in 2014, predating the preferred five-year window. Relies on pre-captured tracking coordinates rather than extracting position data from raw broadcast video, limiting applicability to direct video analysis.

2.7 FieldMOT: A Field-Registered Multi-Object Tracking (Chen, Gao & Zhang)

Objective: Address tracking instability during frequent camera switches and rotations endemic to sports broadcasts by performing tracking in 2D field space rather than image space.

Methodology: “Register-then-track” paradigm: Grounding DINO serves as an open-vocabulary model to identify pitch anchors, which train an offline keypoint detector. During online tracking, detections are projected onto the field template for temporally stable association.

Dataset: Synthesized football data from Google Research Football (GRF) and CARLA street-view environments.

Key Results: Significantly improved tracking stability, increasing IDF1 from 26.3% to 30.7% on soccer data during camera-switching events. MOTA reached 0.8048.

Limitations: Assumes a perfectly flat field, causing errors on pitches with topographic variation. Projection errors amplify for objects distant from the camera or in deep occlusion.

2.8 Sports Camera Calibration via Synthetic Data (Chen & Little)

Objective: Develop an automatic single-image camera calibration method leveraging synthetic data to handle diverse stadium environments and variable broadcasting setups.

Methodology: A parameterized camera pose engine generates synthetic edge images by sampling pan, tilt, and focal length. A Siamese network learns discriminative feature descriptors. A dual-GAN architecture segments the pitch and detects field markings in real broadcast images.

Dataset: World Cup 2014 footage and a volleyball action recognition dataset.

Key Results: Achieved 94.5% mean IoU-part on WC14, surpassing the prior DSM baseline. Volleyball evaluation yielded 97.6% IoU, confirming generalization beyond football.

Limitations: Requires human annotations for GAN training, reducing full-automation claims. Accuracy is compromised under heavy shadows or incomplete pitch line visibility. Publication predates the five-year recency cutoff (2019), placing it outside the preferred window.

2.9 No Bells, Just Whistles: Leveraging Geometric Properties (Gutiérrez-Pérez & Agudo)

Objective: Present a minimalist, geometry-centric 3D calibration pipeline that prioritizes speed and simplicity over complex refinement strategies or temporal consistency modules.

Methodology: HRNetV2 predicts heatmaps for a robust grid of 2D and 3D keypoints. The Direct Linear Transform (DLT) algorithm with RANSAC-based outlier rejection recovers the final projection matrix without further smoothing or optimization.

Dataset: SoccerNet-Calibration, World Cup 2014, and TS-WorldCup datasets.

Key Results: Single-view model achieved 80.4% Acc@5 on WC14, outperforming more complex frameworks such as TVCalib. Established the first public multi-view broadcast benchmark on SN-Calib.

Limitations: Requires at least four visible keypoints to function. Accuracy degrades severely under extreme lens distortion (fisheye shots), where the linear geometric model breaks down.

2.10 Automatic Team Assignment and Jersey Number Recognition (Alhejaily et al.)

Objective: Eliminate labor-intensive manual annotation of player identity and performance metrics by automating pitch segmentation, detection, team assignment, and jersey number recognition.

Methodology: HSV-based pitch segmentation; YOLO-based person detection; unsupervised team assignment using Convolutional Autoencoders (CAE) coupled with K -means clustering; transfer learning for jersey number classification on a custom YOLOv5 detector.

Dataset: Authors contributed a new jersey number dataset with 6,500 images covering digits 0–99 under diverse occlusion and motion blur conditions.

Key Results: Ball detection and jersey number recognition both achieved 99% accuracy, substantially outperforming standard OCR methods. Team assignment accuracy reached 98%.

Limitations: Jersey number recognition remains prone to failure under motion blur and severe occlusion during fast play. The tight bounding boxes required for jersey recognition can miss partial or obstructed digits.

2.11 Tracking Futsal Players with a Wide-Angle Lens Camera (Vieira et al.)

Objective: Assess the feasibility of low-cost, single-camera tracking systems for player localization on smaller futsal courts, incorporating robust radial distortion correction.

Methodology: Improved Hough transform algorithm combined with a one-parameter division model extracts and rectifies court lines. Non-linear optimization minimizes the distance between corrected points and their respective line primitives.

Dataset: Official professional futsal game recorded with GoPro Hero3+ wide-angle camera.

Key Results: Reconstruction uncertainty for static court points ranged from 0.07 m to 0.44 m. Automatic frame-by-frame tracking succeeded for approximately 89% of processed frames.

Limitations: Published in 2017, beyond the five-year recency window. Methodology emphasizes geometric line correction with limited modern association algorithms compared

to tracking-by-detection paradigms.

2.12 AI-Driven Offside and Foul Detection Using Computer Vision (Zhang, Yu & Yan)

Objective: Automate real-time detection of offside positions and foul events in football matches by combining deep learning object detection with motion-based trajectory analysis.

Methodology: A hybrid deep learning framework employs YOLOv4 for player and ball detection with low computational overhead. SORT (Simple Online and Realtime Tracking) combined with Kalman filtering maintains inter-frame trajectories. Motion analysis over tracked paths classifies candidate events as offsides or fouls.

Dataset: Football match broadcast footage evaluated across multiple match scenarios with varied stadium environments and lighting conditions.

Key Results: The system achieves real-time event classification by tightly coupling detection and tracking modules. Demonstrated reliable offside line estimation and foul trajectory classification across diverse broadcast conditions.

Limitations: Reliance on YOLOv4 represents an older detection generation compared to more recent architectures (YOLOv8, RT-DETR). SORT’s simplistic association is susceptible to identity switches during fast occlusion, a known weakness relative to modern trackers such as ByteTrack.

2.13 Positioning Heatmap Generation in Soccer Videos (Tanapatiboon et al.)

Objective: Benchmark and compare different YOLO architectures to identify the most accurate model for generating tactical positioning heatmaps from broadcast video.

Methodology: Systematic evaluation of YOLOv5m6, YOLOv5l-tph (Transformer Prediction Head), and YOLOv5l-tph-plus. Detections are projected from image pixels to a top-down pitch template via homographic transformation for final tactical visualization.

Dataset: SoccerTrack dataset featuring both fish-eye and drone-based top-down views, semi-automatically annotated with GPS ground truth.

Key Results: YOLOv5l-tph achieved the highest precision at 0.9868, substantially outperforming the base YOLOv5m6. Generated heatmaps successfully revealed formation spread differences between competing teams.

Limitations: Model exhibits goalkeeper misclassification errors, though authors note goalkeepers are often excluded from strategic heatmap analysis.

2.14 Real Time Football Match Analysis System (Gupta)

Objective: Develop a comprehensive end-to-end computer vision pipeline for real-time football player tracking and ball possession analysis, delivering robust and interpretable insights into match dynamics.

Methodology: Integrates YOLOv8 for object detection and ByteTrack for multi-object tracking. Custom modules handle view transformation, team classification via visual fea-

tures, and possession estimation based on player-ball proximity. Preprocessing includes frame resizing and contrast enhancement.

Dataset: Open-source annotated football footage from Roboflow and Kaggle platforms.

Key Results: System delivers interpretable match insights by combining state-of-the-art detection and tracking models. End-to-end functionality demonstrated on diverse broadcast sources.

Limitations: Documentation emphasizes architectural design with limited performance benchmarking across diverse stadium environments and extreme lighting. System performance inherently depends on training dataset diversity and annotation quality.

3 Comparative Analysis

ID	Objective	Methodology	Primary Dataset	Main Results	Specific Limitations
1	Assess player/team contributions and spatial dominance using Voronoi diagrams.	Partitions pitch into polygons representing player control zones; calculates “excess Voronoi area”.	Simulated computer soccer game.	Dominant area ratio: 4.3 during goals; mean vertices: 5.2–6.1.	Simulated data only; exceeds 5-year recency (2004).
2	Recover 3D camera parameters from broadcast frames for field registration.	Differentiable objective function minimizing segment reprojection loss; DeepLabV3 segmentation.	SoccerNet-Calib, WC14.	CS: 76.9 (SN-Calib GT); Acc@5: 68.7%.	Sensitive to local minima; performance drops with predicted segmentations.
3	Address MOT challenges in sports (fast motion, similar appearances).	MixSort framework integrating MixFormer structure into tracking-by-detection.	SportsMOT (240 sequences).	HOTA: 74.1; IDF1: 74.4; MOTA: 96.5.	Kalman filters poor on variable-speed motion; needs specialized detectors.
4	Improve calibration by exploiting structural keypoints and intersections.	HRNetV2 detects 57 keypoints; “Iterative Voter” selects reliable subsets.	SoccerNet Challenge 2023.	Acc@5: 0.7322; Final Score: 0.5535 (Winner).	Lower accuracy on unmarked points; radial distortion optimization difficult.
5	All-in-one system for MOT, tactical localization, perspective augmentation.	YOLOv7 backbone; Hungarian algorithm for identity; activation maps for homography.	SoccerNet, ISSIA, NRK.	mAP: 95.7%; MOTA: 94.04%; IDF1: ~93%.	No real-time guarantee; IDSW during full occlusions; wide-angle distortion.
6	Characterize team “styles” using tracking data and formation descriptors.	LDA processes match descriptors; k -NN classifier predicts team identity.	Prozone (20 teams, 375 games).	CCR: 70.38% (fused descriptors).	Outside 5-year window (2014); uses pre-captured data not raw video.
7	Maintain tracking stability across camera switches/rotations via field space.	“Register-then-track” paradigm; Grounding DINO for anchors; field projection.	GRF (synthetic), CARLA.	IDF1: 30.7% (soccer); MOTA: 0.8048.	Assumes flat field; projection errors increase for distant objects.

ID	Objective	Methodology	Primary Dataset	Main Results	Specific Limitations
8	Calibrate cameras from single images using synthetic data.	Sampling engine generates synthetic edges; Siamese network learns descriptors; two-GAN.	WC14, Volleyball.	Mean IoU-part: 94.5% (WC14).	Requires human annotation for GAN; fails under heavy shadows; exceeds 5-year recency (2019).
9	Minimalist geometry-based 3D calibration without refinement modules.	HRNetV2 predicts keypoint heatmaps; DLT algorithm with RANSAC.	SN-Calib, WC14, TS-WorldCup.	Acc@5: 80.4% (WC14); Final Score: 0.571 (SN23).	Requires ≥ 4 visible keypoints; fails under extreme fisheye distortion.
10	Automate pitch segmentation, detection, team assignment, jersey recognition.	YOLO detection; CAE+K-means for team assignment; transfer learning for jersey.	Custom Jersey Dataset (6,500).	mAP: 99% (ball); Accuracy: 99% (jersey).	Jersey recognition hindered by motion blur and occlusions.
11	Assess radial distortion correction for low-cost camera tracking.	Improved Hough transform with division model; nonlinear optimization.	GoPro Hero3+ futsal footage.	Tracking success: 89% of frames.	Published 2017 (exceeds 5-year); geometric focus lacks deep learning.
12	Automate real-time offside and foul event detection in football matches.	YOLOv4 detection; SORT + Kalman filters for tracking; motion analysis for event classification.	Football broadcast footage (multi-match).	Real-time offside estimation and foul classification across diverse conditions.	YOLOv4 is an older detection generation; SORT susceptible to IDSW during fast occlusion.
13	Compare YOLO models for tactical positioning heatmaps.	Evaluates YOLOv5 variants; homographic projection to pitch template.	SoccerTrack (fish-eye, drone).	Precision: 0.9868 (YOLOv5l-tph).	Goalkeeper misclassification (though often excluded).
14	End-to-end pipeline for player tracking and possession analysis.	YOLOv8 detection; ByteTrack tracking; K-means team classification.	Roboflow, Kaggle open-source.	mAP: 91.2% (players); MOTA: 87.4%.	Color similarity confuses teams; ball tracking lags during blur.

4 Gap Analysis

Current academic and commercial literature reveals significant fragmentation across computer vision tasks for football video analysis. While state-of-the-art tracking frameworks such as MixSort [3] and SportsMOT achieve high accuracy in the detection-and-tracking phase, they typically terminate at raw trajectory output without advancing toward actionable tactical intelligence. Conversely, specialized research into 3D camera calibration [2, 9] delivers precise field registration but operates as a standalone component disconnected from downstream analysis modules.

This project addresses multiple critical research gaps:

4.1 Integration of Individual and Collective Analytics

Existing work predominantly emphasizes either tracking-by-detection [3] or aggregate tactical heatmaps [13]. This system bridges that gap by layering both single-player performance profiling and team formation mapping on top of a shared computer vision infrastructure, eliminating redundant computation and enabling cohesive cross-player and team-level insights.

4.2 Automated Formation Labeling from Raw Video

While prior research [6] explored team style analysis, such work relied on pre-processed tracking data supplied externally. This project automates formation discovery by implementing sliding-window averaging and an artificial neural network classifier to generate real-time formation strings (4-3-3, 4-4-2, etc.) directly from raw broadcast video, without manual data preparation.

4.3 Granular Player Performance Profiling

Consumer-level sports analytics tools seldom offer integrated metrics such as touch detection, which this project implements by computing player-ball proximity in real-world coordinates. While systems such as Zhang et al. [12] automate event-level detection (offside, fouls), this project extends beyond binary event flagging [14] to deliver a comprehensive per-frame analytical profile for the selected target player.

4.4 High-Precision Homography Calibration

This project leverages a fine-tuned YOLOv8 model detecting up to 32 pitch landmarks to construct a high-precision homography matrix, enabling physically grounded speed and distance calculations in real-world metre coordinates rather than pixel space. This stands in contrast to prior pipelines that either omit calibration entirely or rely on coarser line-fitting approaches without explicit landmark regression.

4.5 Democratization and Accessibility

While professional systems exist (Prozone, InStat, StatsBomb), they remain prohibitively expensive and proprietary. This project delivers a software-only pipeline optimized for standard YouTube footage and tactical camera feeds, making automated match analysis accessible to smaller clubs, academies, and independent researchers.

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AI Tool Disclosure

I used the help of Claude to help me structure this report, rewrite some sentences in a more professional format and with structuring citations properly. Otherwise, all findings and visualisations are me and my team’s own work.